SOIL EROSION AND CONSERVATION · NOTE CM08

Agroforestry, Grassland Management and Afforestation for Soil Conservation

Course topics: SWCE8.15 L3, SWCE7.4 L9

Agroforestry

Agroforestry develops integrated, self-sustaining land-use systems in which trees and shrubs — fruit, timber and fodder trees, bamboos, cane, palms — grow on farmland with field crops, introduced or retained together or in sequence, often with pasture. It aims to conserve and improve land, optimise the combined output of trees and crops, sustain productivity and give high returns.

System	Components and practices
Agri-silviculture	Agricultural (including fodder) crops with fuel and fodder trees and shrubs — alley cropping, shelterbelts and windbreaks, fuel-wood lots, multipurpose trees such as subabul (*Leucaena leucocephala*)
Silvi-pastoral	Fuel and fodder trees with pasture for livestock — trees on pastures and rangelands, plantation crops with pasture and livestock, vegetative fences, fodder banks
Agri-silvi-pastoral	Crops, fuel/fodder trees and grasses or forage for livestock — home gardens with animals, alley cropping, woody hedgerows round homesteads
Agri-horti-pastoral	Fruit and vegetable crops, field crops and fuel/fodder trees
Multipurpose forest-tree production	Forest species grown and managed on farms for timber, fodder and sometimes food

Agri-horticulture-based systems

- Grow horticultural and food crops on the same land for higher productivity, using inter-crops, under-planting, hedgerow planting and alley cropping.
- **Agri-horticulture:** crops between rows of fruit trees spaced 5–7 m — seasonal income from crops, long-term returns from fruit. Turmeric and ginger tolerate shade; maize, sorghum and groundnut do not. Other inter-crops: coffee, cocoa, banana, amorphophallus, beans.
- **Agri-silvi-horticulture:** crops with top-feed and fruit trees for food, fruit and fodder. **Silvi-pastoral (fruit-based):** thin-canopied, high-bearing fruit trees with top-feed trees for fuel and fodder. **Horti-pastoral:** fruit trees with high-quality fodder grasses and legumes to boost animal husbandry.

Grassland management

- Producing forage from natural vegetation (mostly grasses and herbs) on non-arable land, used by grazing in situ, stall feeding or both.
- Grazing lands cover 80 million ha in India, and overgrazing causes heavy erosion. Grass cuts runoff and erosion, improves aggregation and structure and raises water-holding capacity. In Jharkhand, soil loss was 620 kg/ha from grassland against 2780 kg/ha from overgrazed land.
- **Recommendation:** 25–30% of non-arable land under permanent grass, and land of 8–18% slope under permanent pasture.

Improvement	Practice
Closure to grazing	Complete or rotational for 1–5 years; fencing may be needed, but social fencing is more effective and economical
Soil and water conservation	Light harrowing, contour trenching, furrowing, gully control, water harvesting and spreading
Weed control	Grub out troublesome weeds; weedicides are expensive and may harm livestock; controlled burning on heavily infested grassland
Legumes	*Dolichos lablab*, *D. axillaris*, *Atylosia scarabaeoides*, *Clitoria ternatea*, *Rhynchosia minima*, *Pueraria hirsuta* (kudzu), *Sesbania aegyptiaca*, *Stylosanthes humilis*, *S. hamata*, *Phaseolus trilobus*; kharif pulses mung (*Phaseolus aureus*), moth (*P. aconitifolius*), guar (*Cyamopsis tetragonoloba*)
Fertilisers	20–40 kg N/ha benefits perennial grasses when fertility cannot be kept up naturally
Reseeding and planting	Rooted slips succeed better. Grasses: *Dichanthium annulatum*, *Sehima nervosum*, *Panicum antidotale*, *P. maximum*, *Cenchrus ciliaris*, *C. setigerus*, *Chrysopogon fulvus*

- **Planting conditions:** up to 1500 m elevation, during the monsoon; contour furrows 70 cm apart and 10–15 cm deep on slopes ≤ 33%, or 30 cm deep trenches on slopes > 33% up to 60%. Rooted slips 50–70 cm apart; on flat land 50 × 50 or 75 × 75 cm.
- **Grazing systems:** controlled grazing by carrying capacity (year-long or seasonal); deferred grazing (complete, early or late deferment) to let plants regenerate; rotational grazing in paddocks with rest periods; deferred-cum-rotational grazing combining both.
- **Top feeds:** *Acacia nilotica*, *Albizia lebbeck*, *Ailanthus excelsa*, *Dalbergia sissoo*, *Azadirachta indica*, *Morus alba*, *Moringa oleifera*, *Anogeissus pendula*, *A. latifolia*, *Terminalia tomentosa*, *Ougeinia oojeinensis*, *Bauhinia* spp.
- **Silage:** store rainy-season forage for winter and summer.

Afforestation

- Forests protect soil at canopy, ground and root level — absorbing raindrop energy, reducing wind and water erosion and moderating the microclimate. Keep permanent forest on slopes > 15%.
- About 19% of India's land is under forest, against a target of 30%.
- **Afforestation** raises forest crops on land not already forested — for timber, fuel-wood, fibre, fruit, environmental protection and soil and water conservation — preferring fast-growing, top-feed, drought-resistant, locally adapted species.

Updated figures: the India State of Forest Report 2023 gives forest cover of 21.76% of the geographical area (25.17% with tree cover), and the National Forest Policy 1988 sets the goal at one-third (33%) of the land — two-thirds in the hills.

Procedure

- **Advance plan:** area description, objectives, climate data, stock map of vegetation, suitable species, seed and seedling needs, nursery availability, cost estimate. **Site description:** aspect, slope, elevation, weeds, drainage, rock outcrops. **Site preparation:** clear woody bushes.
- **Species:** by climate and location, coppicing species suited to community needs. Coppicing power depends on species and stump age — good coppicers: *Tectona grandis*, *Shorea robusta*, *Acacia catechu*, *Albizia* spp., *Anogeissus* spp., *Eucalyptus* spp., *Syzygium cumini*, *Salix* spp., *Prosopis juliflora*.

Soil working	Specification
Pit planting	Pits 45 × 45 × 45 cm; 45–50 cm deep in normal soils, 75–100 cm in shallow or gravelly soils
Trench/ridge	1.5 m × 0.45 m × 0.45 m, staggered on the contour, spoil downhill
Mound planting	60 cm high, in waterlogged areas; 1–2 kg gypsum if saline
Refilling pits	Expose the soil for 3–4 weeks before the rains

- **Fencing:** barbed wire or bio-fencing. **Seedlings:** long, well-developed, untwisted roots; healthy and vigorous.
- **Planting:** after a soaking rain with soil moist to 30 cm; firm the soil; keep 15% extra seedlings for replacement. Weed early (essential in the first year) and replace casualties. First harvest in the third or fourth year.
- **Management:** protect from fire, control grazing, fell scientifically, introduce better species; contour bunding and trenching help.

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. Forest Survey of India, India State of Forest Report 2023; National Forest Policy, 1988. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Fruit trees in an agri-horticulture system are spaced about

(a) 1–2 m (b) 5–7 m
(c) 15–20 m (d) 30 m

Q2. Which crop tolerates shade under fruit trees?

(a) maize (b) sorghum
(c) groundnut (d) turmeric

Q3. Grazing lands in India cover about

(a) 8 Mha (b) 30 Mha
(c) 80 Mha (d) 180 Mha

Q4. Land of what slope should be kept under permanent pasture?

(a) 0–3% (b) 3–8%
(c) 8–18% (d) > 33%

Q5. Closure of grassland to grazing is usually for

(a) 1–3 months (b) 1–5 years
(c) 10 years (d) 25 years

Q6. Pit size for afforestation is typically

(a) 30 × 30 × 30 cm (b) 45 × 45 × 45 cm
(c) 1 × 1 × 1 m (d) 2 × 2 × 1 m

Q7. Extra seedlings kept for replacement are about

(a) 5% (b) 15%
(c) 30% (d) 50%

Q8. Permanent forests are recommended on slopes steeper than

(a) 5% (b) 15%
(c) 33% (d) 50%

Q9. The National Forest Policy (1988) target for forest cover is

(a) 19% (b) 25%

(c) 33% (d) 50%

Answers

Q1 — (b) 5–7 m

Q2 — (d) turmeric

Q3 — (c) 80 Mha

Q4 — (c) 8–18%

Q5 — (b) 1–5 years

Q6 — (b) 45 × 45 × 45 cm

Q7 — (b) 15%

Q8 — (b) 15%

Q9 — (c) 33%